Boltzmann equation for Compton scattering

$$\begin{cases} \vec{p} + \vec{q} = \vec{p}\,' + \vec{q}\,' \\ \varepsilon(p) + \varepsilon(q) = \varepsilon(p') + \varepsilon(q') \end{cases}$$

$\vec{p}, \vec{p}\,'$ momenta of electrons
$\vec{q}, \vec{q}\,'$ of photons

Introduce distribution function $f(t, \vec{x}, \vec{p}) = f$
Number of particles in dV of the gas, with momenta within $d^3\vec{p}$ of $\vec{p}$: $\dfrac{dV\, d^3\vec{p}\, f}{(2\pi\hbar)^3}$

$$\frac{\partial f}{\partial t} + \vec{v}(p) \cdot \vec{\nabla}_x f = I[f]$$

$\vec{v}(p)$: due to driving force

$I[f]$: collision integral

"scattering in"

$$I[f(p)] = \iiint W(p, q; p', q') \Big[f(p') f(q') \big(1 + \tau f(p)\big)\big(1 + \tau f(q)\big)$$
$$- f(p) f(q) \big(1 + \tau f(p')\big)\big(1 + \tau f(q')\big) \Big] d^3q\, d^3p'\, d^3q'$$

"scattering out"

Scattering OUT

Particle with momentum $\vec{p}$ will be scattered during dt by some particle in the element $d\vec{q}$, with the end result that the two particles end up in elements $d\vec{p}\,', d\vec{q}\,'$.
This depends on number of particles in $d\vec{q}$, $f(\vec{q})\, d\vec{q}$.
The number of such collisions: $f(\vec{p})\, \cancel{d}\, [f(q)\, d^3\vec{q}]$.

But the end states, $\vec{p}\,'$ and $\vec{q}\,'$, have to be <u>empty</u> so that our particles can transition there $\Rightarrow$

obtain $\left(1+\tau f(p')\right)\left(1+\tau f(q')\right)$

Here $\begin{cases} \tau = -1 & \text{for fermions} \\ \tau = +1 & \text{for bosons} \end{cases}$.

Similar process for "scattering in".

Now, we have fermions $\left[\vec{p} \to \vec{p}\,'\right]$ and bosons (photons) $\left[\vec{q} \to \vec{q}\,'\right]$

$$\Rightarrow \begin{cases} f_e(p')\, f_\gamma(q')\left(1-f_e(p)\right)\left(1+f_\gamma(q)\right) & \text{"IN"} \\ f_e(p)\, f_\gamma(q)\left(1-f_e(p')\right)\left(1+f_\gamma(q')\right) & \text{"OUT"} \end{cases}$$

Electron gas is considered to be dilute and non-degenerate, which means it is described by Maxwell's distribution at temperature T_e.

$$f_e(\vec{p}) = \frac{n_e}{\left(2\pi m k_B T_e\right)^{3/2}}\, e^{-\frac{p^2}{2m k_B T_e}} \qquad \varepsilon(p) = \frac{p^2}{2m_e}$$

Note $f_e \ll 1 \Rightarrow 1 - f_e(p) \approx 1$, $1 - f_e(p') \approx 1$.

Next denote photon distribution function by $n(\omega)$

$\omega = cq$

"IN" $f_e(p')\, n(q')\left(1+n(q)\right)$

"OUT" $f_e(p)\, n(q)\left(1+n(q')\right)$

Then arrive at Eq. (7.48) of the textbook:

$$\frac{\partial n(\omega)}{\partial t} = c \int d^3\vec{p} \int \underbrace{\frac{d\sigma}{d\Omega}}_{\substack{\text{Thomson} \\ \text{scattering} \\ \text{crosssection}}} d\Omega \left\{ f_e(\vec{p}_1)\, n(\omega_1)\left(1+n(\omega)\right) - f_e(\vec{p})\, n(\omega)\left(1+n(\omega_1)\right) \right\}$$

Scattering process: $p + \omega \rightleftarrows p_1 + \omega_1$.

Observe that it conserves number of photons (and electrons)

$$\frac{d\sigma}{d\Omega} = \frac{1}{2} r_0^2 \left(1+\cos^2\theta\right) \quad \text{where } \theta = \text{scattering angle.}$$

Comment on the origin of $\underline{1+n}$ factor — stimulated emission!

$$\left.\begin{array}{ll} \text{creation} & a^+ |n\rangle = \sqrt{1+n}\, |n+1\rangle \\ \text{annihilation} & a\, |n\rangle = \sqrt{n}\, |n-1\rangle \end{array}\right\} \text{for bosons}$$

While for fermions $n = 0, 1$ (Pauli principle)

$$\Rightarrow \quad a^+|0\rangle = |1\rangle \quad \Rightarrow \quad a^+|n\rangle = \sqrt{1-n}\, |n+1\rangle$$
$$a^+|1\rangle = 0$$

Fokker-Planck eq. for $n(\omega)$

$$\frac{\partial n(\omega)}{\partial t} = c \int d^3p \int d\Omega \, \frac{d\sigma}{d\Omega} \Big[f_e(\bar{p}_1)\, n(\omega_1)\big(1+n(\omega)\big) -$$

$$- f_e(\bar{p})\, n(\omega)\big(1+n(\omega_1)\big)\Big]$$

$$\varepsilon(\bar{p}) + \hbar\omega = \varepsilon(\bar{p}_1) + \hbar\omega_1$$

Energy transfer to the photons $\Delta = \frac{\hbar(\omega_1-\omega)}{k_B T} \ll 1$.

$\Rightarrow$ expand distribution function

$$n(\omega_1) = n(\omega + \omega_1 - \omega) \simeq n(\omega) + (\omega_1-\omega)\frac{\partial n(\omega)}{\partial \omega} + \frac{1}{2}(\omega_1-\omega)^2 \cdot \frac{\partial^2 n}{\partial \omega^2}$$

let $x = \frac{\hbar\omega}{k_B T} \Rightarrow \frac{\partial}{\partial \omega} = \frac{\partial}{\partial x}\frac{\partial x}{\partial \omega} = \frac{\hbar}{k_B T}\frac{\partial}{\partial x}$

$$\Rightarrow n(\omega_1) \simeq n(\omega) + \Delta \frac{\partial n}{\partial x} + \frac{1}{2}\Delta^2 \frac{\partial^2 n}{\partial x^2} + \dots$$

Then $$\frac{\partial n(\omega)}{\partial t} = c \int d^3\vec{p}\, \frac{d\sigma}{d\Omega}\Big\{ f(\bar{p}_1)\big(n + \Delta n' + \tfrac{1}{2}\Delta^2 n''\big)(1+n)$$

$$- f(\vec{p})\, n\big(1 + n + \Delta n' + \tfrac{1}{2}\Delta^2 n''\big)\Big\} =$$

$$= c \int d^3\vec{p}\, \frac{d\sigma}{d\Omega}\Big\{ f(\bar{p}_1)\big[n(1+n) + \Delta n'(1+n) + \tfrac{1}{2}\Delta^2 n''(1+n)\big]$$

$$- f(\vec{p})\big[n(1+n) + \Delta n' n + \tfrac{1}{2}\Delta^2 n'' n\big]\Big\}$$

<u>Note</u>: electrons obey Maxwell's distribution.

$$f(\bar{p}_1) \equiv f(\varepsilon(\bar{p}_1)) = n_e (2\pi m_e k_B T)^{-3/2} e^{-\frac{\varepsilon(p_1)}{k_B T}}$$

$$e^{-\frac{\varepsilon(p_1)}{k_B T}} = e^{-\frac{\varepsilon(p) + \hbar(\omega-\omega_1)}{k_B T}} = e^{-\frac{\varepsilon(p)}{k_B T}} e^{\Delta} \approx e^{-\frac{\varepsilon(p)}{k_B T}}\big(1 + \Delta + \tfrac{1}{2}\Delta^2 + \dots\big)$$

$$\Rightarrow f(\bar{p}_1) = f(\bar{p})\big(1 + \Delta + \tfrac{1}{2}\Delta^2\big)$$

$$\frac{1}{c}\frac{\partial n(\omega)}{\partial t} = \int d^3\vec{p}\, \frac{d\sigma}{d\Omega} f(\vec{p}) \Big\{ \left(1+\Delta+\tfrac{1}{2}\Delta^2\right)\left[n(1+n) + \Delta n'(1+n) + \tfrac{1}{2}\Delta^2 n''(1+n)\right]$$

$$- \left[n(1+n) + \Delta n' n + \tfrac{1}{2}\Delta^2 n'' n \right] \Big\} =$$

$$= \int d^3\vec{p}\, \frac{d\sigma}{d\Omega} f(\vec{p}) \Big\{ \left(\Delta + \tfrac{1}{2}\Delta^2\right) n(1+n) + \left(1+\Delta+\tfrac{1}{2}\Delta^2\right)\Delta n'(1+n)$$

$$- \Delta n' n + \left(1+\Delta+\tfrac{1}{2}\Delta^2\right)\cdot \tfrac{1}{2}\Delta^2 n''(1+n) - \tfrac{1}{2}\Delta^2 n'' n \Big\}$$

Collect terms to Δ^2 accuracy:

$$\frac{1}{c}\frac{\partial n}{\partial t} = \int d^3\vec{p}\, \frac{d\sigma}{d\Omega} f(\vec{p}) \Big\{ \left(\Delta + \tfrac{1}{2}\Delta^2\right) n(1+n) + (1+\Delta)\Delta n'(n+1)$$

$$- \Delta n' n + \tfrac{1}{2}\Delta^2 n''(1+n) - \tfrac{1}{2}\Delta^2 n'' n \Big\} =$$

$$= \left(\int d^3\vec{p}\, \frac{d\sigma}{d\Omega} f(\vec{p})\, \Delta \right) \left[n(1+n) + n'(n+1) - n' n \right]$$

$$+ \left(\int d^3\vec{p}\, \frac{d\sigma}{d\Omega} f(\vec{p})\, \Delta^2 \right) \left[\tfrac{1}{2} n(1+n) + n'(n+1) + \tfrac{1}{2} n'' \right] + O(\Delta^3).$$

$\Rightarrow$ finally $$\frac{1}{c}\frac{\partial n}{\partial t} = \left[n' + n(1+n) \right] I_1 + \left[\tfrac{1}{2} n'' + n'(n+1) + \tfrac{1}{2} n(1+n) \right] \cdot I_2$$

where $$I_1 = \int d^3\vec{p}\, \frac{d\sigma}{d\Omega} f(\vec{p})\, \Delta$$

This is Eq. (7.52)

$$I_2 = \int d^3\vec{p}\, \frac{d\sigma}{d\Omega} f(\vec{p})\, \Delta^2 .$$

Start by working out I_2. Need to express Δ via $\vec{p}$ (initial momentum of electron) since the integral is over $d^3\vec{p}$.

Kinematics of Compton scattering

$$h\nu\, h\nu_1 (1 - \hat{k}_i \cdot \hat{k}_f) = h\nu (E_i - c\vec{p}_i \cdot \hat{k}_i) - h\nu_1 (E_i - c\vec{p}_i \cdot \hat{k}_f)$$

Now, $h\nu = \hbar\omega$, $h\nu_1 = \hbar\omega_1$; $\hat{k}_i = \hat{n}$ (unit vector along $\vec{k}_i$)

$E_i = E$ (initial energy of electron) $\hat{k}_f = \hat{n}_1$ (along $\vec{k}_f$).

$$\Rightarrow \quad E(\hbar\omega_1 - \hbar\omega) = c\hbar\omega_1\, \vec{p} \cdot \hat{n}_1 - c\hbar\omega\, \vec{p} \cdot \hat{n} + \hbar^2 \omega\omega_1 (1 - \hat{n} \cdot \hat{n}_1);$$

$$\cancel{= c(\hbar\omega_1 - \hbar\omega)\, \vec{p} \cdot (\hat{n}_1 - \hat{n}) + c\hbar\omega\, \vec{p} \cdot (\hat{n}_1 - \hat{n}) + c\hbar\omega\, \vec{p} \cdot \hat{n} - \hbar\omega(\hbar\omega_1 - \hbar\omega + \hbar\omega)(1 - \hat{n} \cdot \hat{n}_1)}$$

$$\hbar\omega_1 = k_B T \cdot \Delta + \hbar\omega = k_B T(\Delta + x) \quad \text{since } \hbar\omega = x k_B T$$

$$\Rightarrow \quad E(k_B T \Delta) = c k_B T(\Delta + x)\, \vec{p} \cdot \hat{n}_1 - c x k_B T\, \vec{p} \cdot \hat{n} - x (k_B T)^2 (\Delta + x)(1 - \hat{n} \cdot \hat{n}_1)$$

Divide by $k_B T$:

$$\Delta E + \underset{k_B T}{x} \Delta (1 - \hat{n} \cdot \hat{n}_1) - \Delta\, c\vec{p} \cdot \hat{n}_1 = c x\, \vec{p} \cdot \hat{n}_1 - c x \vec{p} \cdot \hat{n} - x^2 k_B T (1 - \hat{n} \cdot \hat{n}_1)$$

So
$$\Delta = \frac{x c \vec{p} \cdot (\hat{n}_1 - \hat{n}) - x^2 k_B T (1 - \hat{n} \cdot \hat{n}_1)}{E - c\vec{p} \cdot \hat{n}_1 + x k_B T (1 - \hat{n} \cdot \hat{n}_1)}$$

Now $E \simeq mc^2 \gg cp$ since for Maxwell distribution $p \sim (m k_B T)^{1/2}$.

In addition $x = \frac{\hbar\omega}{k_B T} \ll 1$ by assumption so that $x^2 \ll x \Rightarrow$

$$\Rightarrow \quad \Delta \simeq x\, \frac{\vec{p} \cdot (\hat{n}_1 - \hat{n})}{mc}.$$

$$I_2 = \int d^3\vec{p}\, \frac{d\sigma}{d\Omega}\, \Delta^2 f(\vec{p}) = \int d^3\vec{p}\, \frac{d\sigma}{d\Omega} \left(\frac{x}{mc}\right)^2 \left(\vec{p}\cdot(\hat{n}_1 - \hat{n})\right)^2 \cdot$$

$$\cdot \frac{n_e}{(2\pi m k_B T)^{3/2}}\, e^{-\frac{p^2}{2mk_BT}} \sim \left(\frac{x}{mc}\right)^2 \int dp\, p^2 \cdot \underbrace{p^2}_{f_e(p)} \int d\Omega\, \frac{d\sigma}{d\Omega}$$

$$p = (2mk_BT)^{1/2}\, y$$

$$= \left(\frac{x}{mc}\right)^2 (2mk_BT)^{5/2} \frac{n_e}{(2\pi m k_B T)^{3/2}} \int_0^\infty dy\, y^4 e^{-y^2} \int d\Omega\, \frac{d\sigma}{d\Omega}$$

$$= \left(\frac{x}{mc}\right)^2 n_e (2mk_BT) \underbrace{\int dy\, y^4 e^{-y^2} \int \frac{d\Omega}{\pi^{3/2}} \frac{d\sigma}{d\Omega} \left(\hat{p}\cdot(\hat{n}_1 - \hat{n})\right)^2}_{\text{just a number}\,\cdot\,\sigma_T}$$

$$= x^2 n_e \frac{k_B T}{mc^2} \sigma_T \cdot (2) \quad [\text{Prob. } 7.4]$$

I_1 can be found by an elegant trick due to Kolomeetz. We start by observing that Compton scattering <u>conserves</u> number of photons $e + \nu \to e' + \nu'$.

This is very different from emission (absorption) when number of photons change by $+1$ (-1).

This means that $\int n(\omega)\, d^3\vec{k} = \frac{1}{c^3}\int n(\omega)\,\omega^2 d\omega \int d\Omega =$

$$= \frac{4\pi}{c^3}\int n(\omega)\,\omega^2 d\omega \text{ is constant.}$$

Since $\hbar\omega = x\, k_B T$ it means $\int n(x)\, x^2 dx = \text{const.}$

$$\Rightarrow \quad \frac{d}{dt}\int n\, x^2 dx = \int \frac{\partial n}{\partial t}\, x^2 dx = 0$$

But $\frac{\partial n}{\partial t} + \vec{\nabla}\cdot\vec{j} = 0$ continuity equation for <u>conserved</u> density n, in terms of its current.

Due to assumed spherical symmetry $\vec{\nabla}\cdot\vec{j} = \frac{1}{x^2}\frac{\partial}{\partial x}\left(x^2 j(x)\right)$

$$\Rightarrow \quad \frac{\partial n}{\partial t} = -\frac{1}{x^2}\frac{\partial}{\partial x}\left(x^2 j(x)\right) \text{ so that}$$

$$\int \frac{\partial n}{\partial t}\, x^2 dx = -\int \frac{\partial}{\partial x}\left(x^2 j(x)\right) dx = -x^2 j(x)\Big|_0^\infty = 0$$

Now, $\frac{\partial n}{\partial t} = -\frac{1}{x^2}\left(2x\, j(x) + x^2 j'(x)\right) = -\frac{2}{x}\, j(x) - j'(x)$.

while Fokker-Planck equation reads

$$\frac{\partial n}{\partial t} = \left[n' + n(1+n)\right] c I_1 + \left(\frac{1}{2}n'' + n'(n+1) + \frac{1}{2}n(n+1)\right) c I_2$$

$$= C_1(x)\, n'' + C_2(n,x)\, n' + C_3(n,x)$$

where we made explicit all factors of $n' = \frac{\partial n}{\partial x}$, $n'' = \frac{\partial^2 n}{\partial x^2}$.

Comparing two $\frac{\partial n}{\partial t}$ forms, we conclude that

$$j = g(x)\left[n' + h(n,x)\right]$$

so that j' will contain n'' term with n-independent coefficient.

Next, the current $j(x)$ in ω- (or, in k-) space must vanish for the equilibrium distribution which conserves number of photons.

Such a distribution is Bose-Einstein one:

$$n = \frac{1}{e^{\frac{\hbar\omega-\mu}{k_B T}}-1} = \frac{1}{e^{x+a}-1}, \quad a = -\mu/k_B T.$$

$$\Rightarrow j = 0 \text{ for } n = \frac{1}{e^{x+a}-1} \Rightarrow h(n,x) = -n'(x) =$$

$$= -\frac{d}{dx}\frac{1}{e^{x+a}-1} = +\frac{e^{x+a}}{(e^{x+a}-1)^2} = \frac{e^{x+a}-1+1}{(e^{x+a}-1)^2} =$$

$$= n + n^2 = n(1+n).$$

$$\Rightarrow j(x) = g(x)\left[n' + n(1+n)\right].$$

Thus
$$\frac{\partial n}{\partial t} = -\frac{2g}{x}\left[n' + n(1+n)\right] - g'\left[n' + n(1+n)\right]$$
$$- g\left[n'' + n' + 2n'n\right] =$$
$$= -\left(\frac{2g}{x} + g'\right)\left[n' + n(1+n)\right] - g\left[n'' + n'(1+2n)\right].$$

But again $\frac{\partial n}{\partial t} = \left[n' + n(1+n)\right] c I_1 + \frac{1}{2}\left[n'' + 2n'(1+n) + n(n+1)\right] \cdot$
$\cdot c I_2$

Compare coefficients of n'': $-g = \frac{1}{2} c I_2$.

$$\Rightarrow g = -c x^2 n_e \sigma_T \left(\frac{k_B T}{mc^2}\right).$$

Next, n': $-\left(\frac{2g}{x} + g'\right) - g(1+2n) = c I_1 + (1+n) c I_2$.

$$-\frac{2g}{x} - g' - g + \cancel{(-2g)n} = c I_1 + c I_2 + \cancel{(c I_2)} n$$

$$\Rightarrow c I_1 = -c I_2 - g - g' - \frac{2g}{x}$$

But since $g = -\frac{1}{2} c I_2$, $g' = -\frac{1}{2} c \frac{d}{dx}\left(2x^2 n_e \frac{k_B T}{mc^2} \sigma_T\right) =$

$$= -2cx\, n_e \left(\frac{k_B T}{mc^2}\right) \sigma_T.$$

and $-\frac{2g}{x} = \frac{2}{x}\left(\frac{1}{2} c I_2\right) = \frac{c I_2}{x} = 2x\, n_e \left(\frac{k_B T}{mc^2}\right) \sigma_T c$

$$c I_1 = -c I_2 + \frac{1}{2} c I_2 + 2x\, n_e \left(\frac{k_B T}{mc^2}\right) \sigma_T c + 2x\, n_e \left(\frac{k_B T}{mc^2}\right) \sigma_T c$$

$$= 4x c n_e \left(\frac{k_B T}{mc^2}\right) \sigma_T - \frac{1}{2} c \left(2x^2 n_e \frac{k_B T}{mc^2} \sigma_T\right) =$$

$$= c\, n_e \left(\frac{k_B T}{mc^2}\right) \sigma_T \left(4x - x^2\right)$$

Therefore $I_1 = n_e \sigma_T\, x(4-x) \left(\frac{k_B T}{mc^2}\right)$.

Now we can check ourselves by looking on $n(1+n)$:

$$-\left(\frac{2g}{x} + g'\right) = c I_1 + \frac{1}{2} c I_2$$

But $-\frac{2g}{x} - g' = 2c n_e \sigma_T \frac{k_B T}{mc^2} (x + x)$.

$$c I_1 + \frac{1}{2} c I_2 = c\, n_e \sigma_T \frac{k_B T}{mc^2} \left(4x - x^2 + \frac{1}{2} \cdot 2x^2\right)$$

checks out!

Thus Fokker-Planck eq. is given by

$$\frac{\partial n}{\partial t} = -\frac{1}{x^2}\frac{\partial}{\partial x}\left[x^2 g(x)\left\{n' + n(1+n)\right\}\right]$$

where $g(x) = -c n_e \sigma_T \left(\frac{k_B T}{mc^2}\right) x^2$.

$$\frac{\partial n}{\partial t} = c n_e \sigma_T \left(\frac{k_B T}{mc^2}\right) \frac{1}{x^2}\frac{\partial}{\partial x}\left[x^4 (n' + n + n^2)\right].$$

or, $t_c = n_e \sigma_T c t$

$$\frac{\partial n}{\partial t_c} = \left(\frac{k_B T}{mc^2}\right) \frac{1}{x^2}\frac{\partial}{\partial x}\left[x^4 (n' + n + n^2)\right]. \quad \text{(Eq. 7.59)}$$

Kompaneetz equation (1957).